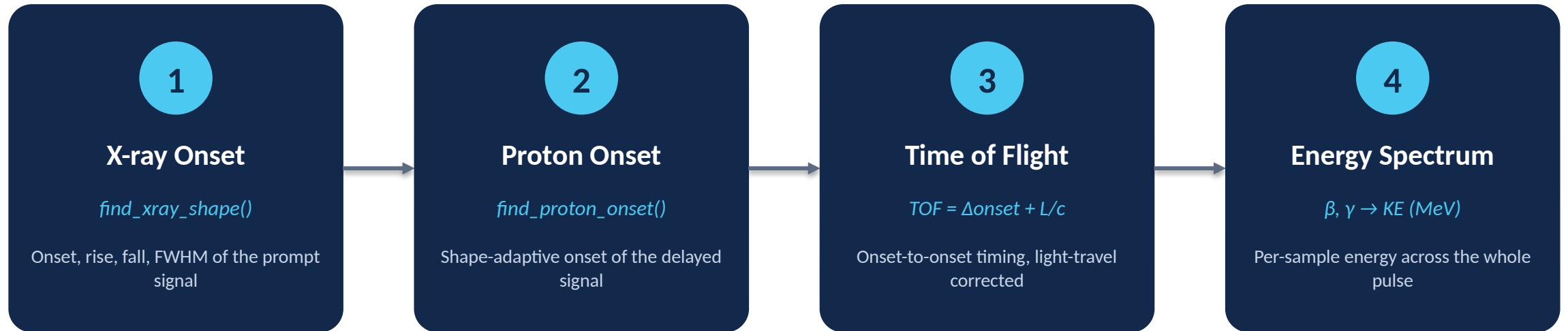
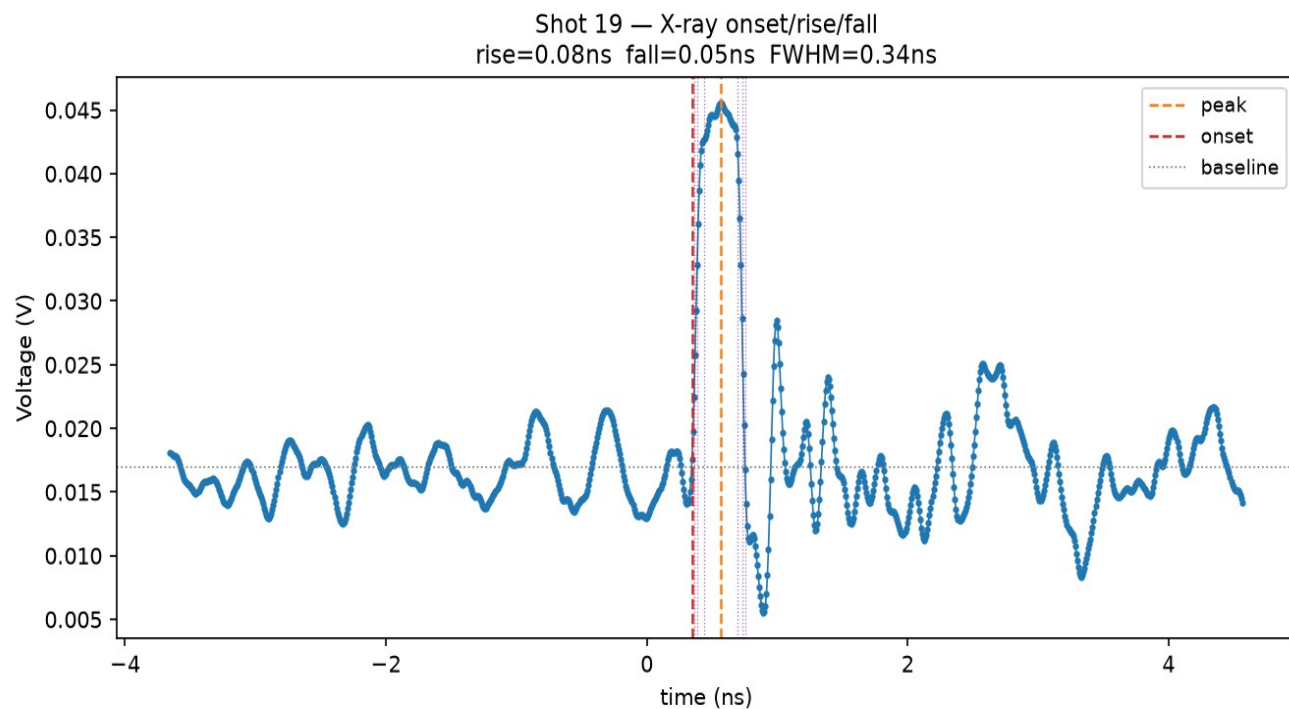


# 1 Analysis Pipeline

Four reusable functions, chained once and shared across every notebook — no duplicated logic.



## 2 X-ray Pulse Characterization



- Onset: walk backward from peak to ~baseline noise ( $2\sigma$ )
- Rise/fall edges bounded to the pulse's own [onset, peak] window
- FWHM from 50% crossing on rise and fall edges
- Validated across all 12 verified double-pulse shots

Rise time **0.08 ns**

Fall time **0.05 ns**

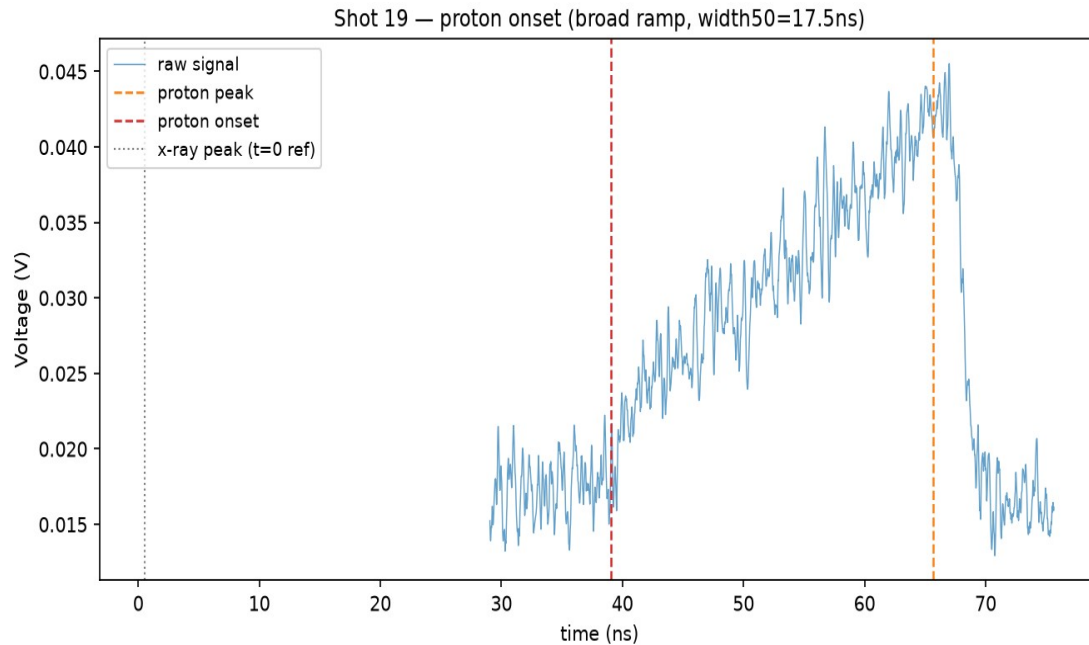
FWHM **0.34 ns**

*Shot 19 shown above*

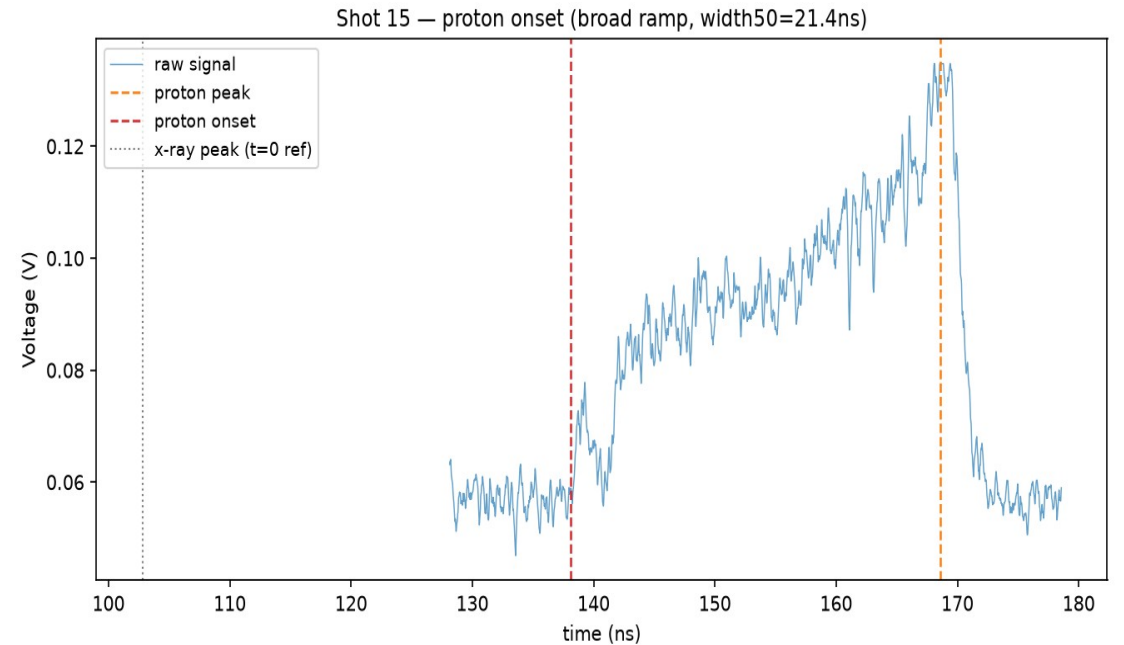
## 3

# Proton Onset Characterization

*Onset threshold adapts to each pulse's measured width — two representative shots from the good, fully-usable set.*



**Shot 19 — rod target**



**Shot 15 — Cu film target**

— Onset threshold set from each pulse's own 50%-width, not a single fixed fraction — validated across the 8 good shots carried through to energy

## 4 Time-of-Flight Results

*Onset-to-onset timing across all 10 usable double-pulse shots.*

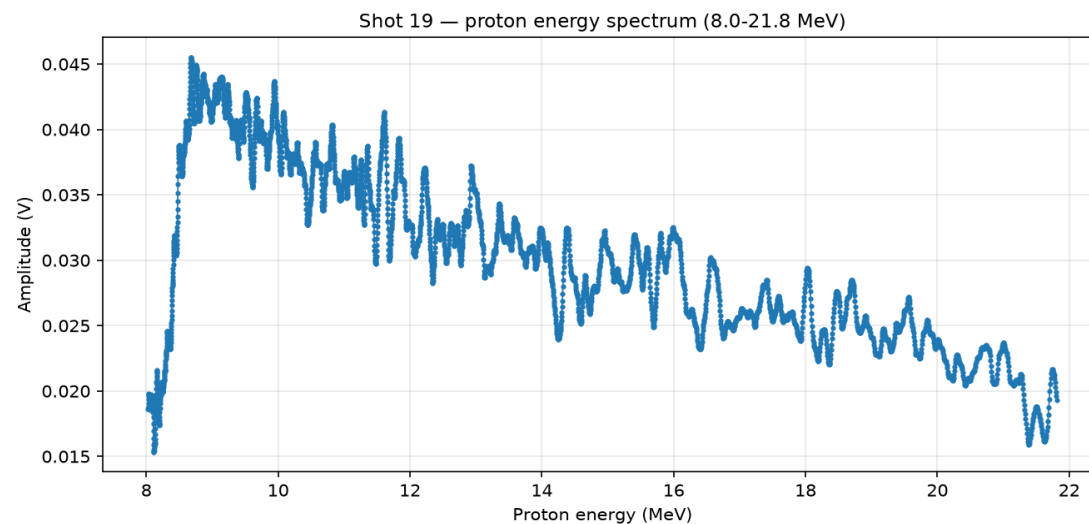
| Shot | Target  | Thickness (μm) | X-ray onset (ns) | Proton onset (ns) | TOF (ns) |
|------|---------|----------------|------------------|-------------------|----------|
| 9    | Cu      | 25             | 96.166           | 108.666           | 12.5     |
| 11   | Al      | 25             | 95.396           | 108.646           | 13.25    |
| 15   | Cu film | 100            | 102.412          | 138.123           | 35.711   |
| 16   | Mylar   | 1              | 102.617          | 131.531           | 28.914   |
| 18   | Al      | 12.5           | 4.576            | 36.217            | 31.641   |
| 19   | rod     | —              | 0.347            | 39.042            | 38.695   |
| 20   | Mylar   | 1              | 1.796            | 28.601            | 26.805   |
| 21   | Mylar   | 1              | 1.758            | 34.516            | 32.758   |
| 22   | Mylar   | 1              | -1.112           | 26.623            | 27.734   |
| 27   | Mylar   | 1              | 1.822            | 36.095            | 34.273   |

## 5 Proton Energy Spectrum

$$\text{TOF} = (t_{\text{proton,onset}} - t_{\text{xray,onset}}) + L/c \rightarrow v = L / \text{TOF} \rightarrow \beta = v / c$$

$$\gamma = 1 / \sqrt{1 - \beta^2} \rightarrow \text{KE} = (\gamma - 1) \cdot m_p \cdot c^2 \text{ if } \beta > 0.1, \text{ else } \text{KE} = \frac{1}{2} \cdot m_p \cdot v^2$$

| Shot | Target  | L (m)   | TOF (ns) | $\beta$ | KE (MeV) |
|------|---------|---------|----------|---------|----------|
| 9    | Cu      | unknown | —        | —       | —        |
| 11   | Al      | unknown | —        | —       | —        |
| 15   | Cu film | 3.12    | 46.118   | 0.2257  | 24.843   |
| 16   | Mylar   | 3.12    | 39.321   | 0.2647  | 34.697   |
| 18   | Al      | 3.12    | 42.048   | 0.2475  | 30.131   |
| 19   | rod     | 3.12    | 49.103   | 0.2119  | 21.812   |
| 20   | Mylar   | 3.12    | 37.212   | 0.2797  | 38.998   |
| 21   | Mylar   | 3.12    | 43.165   | 0.2411  | 28.521   |
| 22   | Mylar   | 3.12    | 38.142   | 0.2729  | 37.007   |
| 27   | Mylar   | 3.12    | 44.681   | 0.2329  | 26.537   |



Shot 19 — representative spectrum, 8.0–21.8 MeV

- Shots 9, 11 excluded: pre-shot-11 flight distance L not yet known
- 8 of 10 shots span roughly 22–39 MeV peak energy

## 6 Conclusions & Next Steps

### What's Solid

- X-ray onset/rise/fall/FWHM validated on all 12 double-pulse shots
- Proton onset shape-adaptive threshold validated on 10 of 12 shots
- Full pipeline connected end-to-end: onset → TOF → energy
- Energy spectra recovered for 8 shots, ~22–39 MeV peak range
- All logic lives in one shared module — no duplicated code across notebooks

### Open Items / Next Steps

- Proton onset detection still broken on shots 25, 26 (bipolar signal shape)
- Pre-shot-11 flight distance  $L$  unknown — blocks energy for shots 9, 11
- Backfill those shots once  $L$  is found and 25/26 are fixed
- Diamond-detector charge/dose calibration not yet integrated